Project of modern physics

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1 Introduction

Thanks to M. Akridas-Morel our referent professor during this project.

This project has be present in the purpose of understanding and modelling a specific case of a potential step, the barrier of potential.

The Python code is available on the GitHub page at the following link :

2 Generality about the Schrödinger Equation

In this part we will remind some important point about the Schrödinger Equation, and waves functions.

2.1 Generality

1. at 3 dimensions, a plane waves is define as :

$$\Psi(\vec{r},t) = Ae^{i(\vec{k}\vec{r}-\omega t)}$$

with $\vec{k}$ the wave vector and ω the natural frequency.

- $[\vec{k}] = [\frac{2\pi}{\lambda}] = L^{-1}$

- $[\omega] = [\frac{2\pi}{T}] = T^{-1}$

2. At 1 dimension :

$$\Psi(x,t) = \Psi_0 e^{i(kx-\omega t)}$$

$$\Re(\Psi(x,t)) = \Psi_0 \cos(kx) \text{ and } \Im(\Psi(x,t)) = \Psi_0 \sin(\omega t)$$

3. To know the dimension of Ψ_0 we begin with the definition of the normalisation.

$$\int_{-\infty}^{+\infty} |\Psi(x,t)|^2 dx = 1$$
$$\Rightarrow [|\Psi_0|^2] * L = 1$$
$$\Rightarrow [\Psi_0] = L^{\frac{-1}{2}} = \frac{1}{\sqrt{m}}$$

4. If the particle is free, V(x) = 0. So with the Schrödinger Equation at 1 dimension :

$$\begin{aligned} -i\hbar\frac{\partial\Psi(x,t)}{\partial t} &= \frac{-\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + V(x)\Psi(x,t) \\ \Rightarrow \hbar\omega\Psi(x,t) &= \frac{\hbar^2 k^2}{2m}\Psi(x,t) \\ \Rightarrow \hbar\omega &= \frac{\hbar^2 k^2}{2m} \end{aligned}$$

5. So with the previous question, we already have :

$$\omega(k) = \frac{\hbar k^2}{2m}$$

so :

$$\begin{aligned} v_\phi &= \frac{\omega(k)}{k} = \frac{\hbar k}{2m} \\ v_g &= \frac{d\omega(k)}{dk} = \frac{\hbar k}{m} \end{aligned}$$

6. let $v = p/m$.
We can notice that, $p = \hbar k$
Indeed, $p = h/\lambda$ and $\lambda = 2\pi/k$
So, $p = hk/2\pi = \hbar k$ Thus, $v = v_g$ and $v = v_\phi/2$

7. The conditions for Ψ to be normalizable, is :

$$I = \int_{-\infty}^{+\infty} |\Psi(x,t)|^2 = 1$$

However, I diverge. So, Ψ isn't normalizable. If the problem was in a infinite barrier of potential, the boundaries would be [0,a].

8. As Ψ aren't normalizable, it's not a proper physical solution.

2.2 Superposition of plane waves

1. The Schrödinger Equation is a order 2 HLDE, so all linear combination of solutions remain solution. So as a superposition of wave functions is a linear combination of wave functions, it remains solution to the Schrödinger Equation.

2. let at t = 0 :

   - $\Psi_1(x,0) = \Psi_1 e^{ikx}$
   - $\Psi_2(x,0) = \frac{1}{2}\Psi_1 e^{i(k+\frac{\Delta k}{2})x}$
   - $\Psi_3(x,0) = \frac{1}{2}\Psi_1 e^{i(k-\frac{\Delta k}{2})x}$

   let :

$$\begin{aligned}\Psi(x,0) &= \Psi_1(x,0) + \Psi_2(x,0) + \Psi_3(x,0) \\ &= \Psi_0 e^{ikx}(1 + \frac{1}{2}e^{i\frac{\Delta k}{2}} + \frac{1}{2}e^{-i\frac{\Delta k}{2}}) \\ &= \Psi_1 e^{ikx}\left(1 + \cos(\frac{\Delta k}{2})\right)\end{aligned}$$

3.

$$\begin{aligned}\Psi(x,0)^*\Psi(x,0) &= \left[|\Psi_1|^2 e^{ikx}e^{-ikx}\left(1+\cos(\frac{\Delta k}{2})\right)^2\right] \\ &= |\Psi_1^2|\left(1+\cos(\frac{\Delta k}{2})\right)^2 \\ &= |\Psi_1|^2\left(1 + 2\cos(\frac{\Delta k}{2}) + \cos^2(\frac{\Delta k}{2})\right) \\ &= \rho(x)\end{aligned}$$

3 Waves packets

3.1 Gaussian waves packets

1. As we said before, $\omega = \frac{\hbar k^2}{2m}$
   So with the solution in the statement, at 1 dimension :

$$\begin{aligned}\Psi(x,t) &= \frac{1}{\sqrt{2\pi}} \int g(k) e^{ikx - i\omega t} dk \\ &= \frac{1}{\sqrt{2\pi}} \int g(k) e^{ikx - i\frac{\hbar}{2m}k^2 t} dk\end{aligned}$$

2.

$$\begin{aligned}\Psi(x,t) &= \frac{1}{\sqrt{2\pi}} \int \frac{\sqrt{a}}{(2\pi)^{\frac{1}{4}}} e^{\frac{-a(k-k_0)^2}{4}} e^{ikx - i\frac{\hbar}{2m}k^2 t} dk \\ &= \frac{\sqrt{a}}{(2\pi)^{\frac{3}{4}}} \int e^{\frac{-a^2}{4}(k-k_0)^2 + 2\frac{a^2}{4}kk_0 + ikx - i\frac{\hbar}{2m}k^2 t} dk \\ &= \frac{\sqrt{a}}{(2\pi)^{\frac{3}{4}}} \int e^{-k^2(\underbrace{\frac{a^2}{4} + i\frac{\hbar}{2m}t}_{A}) + k(\underbrace{\frac{1}{2}a^2 k_0 + ix}_{B}) - \underbrace{\frac{a^2 k_0^2}{4}}_{C}} dk\end{aligned}$$

By applying the standard Gaussian integral identity :

$$I = \int e^{ax^2+bx+c} = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}+c} dx$$

explain Ψ as :

$$\Psi(x,t) = \alpha e^{\beta(k)}$$

We can notice that :

$$4A = a^2 + i\frac{2\hbar}{m}t$$

$$\begin{aligned}\beta(k) &= \frac{\frac{a^4 k_0^4}{4} + ia^2 k_0 x - x^2}{4A} - \frac{a^2 k_0^2}{4} \\ &= \frac{-x^2 + i(a^2 k_0 x - \frac{a^2 k_0^2 \hbar t}{2m})}{4A}\end{aligned}$$

So, in letting :

$$\omega(t) = a\sqrt{1 + \left(\frac{2\hbar t}{ma^2}\right)^2} \text{ and } v_g = \frac{\hbar k}{m}$$

We have,

$$\Psi(x,t) = \underbrace{\left(\frac{2}{\pi\omega(t)^2})\right)^{\frac{1}{4}} e^{i\phi}}_{\Psi_0} e^{\frac{-2(x-v_g t)^2}{\omega(t)^2}}$$

3. In using the previous expression, we can explain the normalization. Let the follow substitution : $u = x - v_g t \Rightarrow \frac{du}{dx} = 1$

$$\begin{aligned}\int_{-\infty}^{+\infty} |\Psi(x,t)|^2 dx &= \int_{-\infty}^{+\infty} |\Psi_0|^2 e^{\frac{-2u^2}{\omega(t)^2}} du \\ &= \sqrt{\frac{2}{\pi}}\frac{1}{\omega} * \omega\sqrt{\frac{\pi}{2}} \\ &= 1\end{aligned}$$

So, $\Psi(x,t)$ is normalize.

4. We will using the Fourier transform, to find $g(k)$, more precisely the invert Fourier transform.

$$\Psi(x,t) = \frac{1}{\sqrt{2\pi}} \int g(k) e^{i(kx-\omega t)} dk$$

$$\Rightarrow \Psi(x,0) = \frac{1}{\sqrt{2\pi}} \int g(k) e^{ikx} dk$$

$$\Rightarrow g(k) = \frac{1}{\sqrt{2\pi}} \int \Psi(x,0) e^{-ikx} dx$$